

Zirui Wang

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WORK EXPERIENCE

System AI Research Engineer

Huawei Technologies R&D (UK) Ltd

Edinburgh, UK

Oct 2025 – Present

- Designed and evaluated adaptive scheduling and KV-cache management policies for concurrent agentic workloads in vLLM-Ascend; increased prefix-cache hit rate from 56% to 72%, halved queue wait, completed batches 17% sooner at equal task-resolve rate, and improved accuracy by 9 points at the highest concurrency tested. Submitted to SoCC 2026.
- Built a cross-layer observability pipeline for agentic LLM serving using OpenTelemetry/OpenLIT, custom trace analyzers, Gantt exporters, and token-level latency profiling; extended vLLM-Ascend to profile KV-cache offloading on Huawei Ascend NPUs and identify coordination and memory bottlenecks.
- Built a reproducible CrewAI experimentation framework to evaluate how agent orchestration topology affects task outcomes, comparing single-agent, hierarchical manager-worker, and decentralized multi-agent designs across GAIA, SWE-bench, and HotpotQA.

Technical Insight Specialist

Huawei Technologies R&D (UK) Ltd

Edinburgh, UK

Nov 2024 – Oct 2025

- Led comparative evaluations of LLM inference across Cerebras wafer-scale, NVIDIA GPU, and Huawei Ascend, analyzing accelerator architecture, memory hierarchy, and interconnect bottlenecks to quantify scale-up versus scale-out trade-offs and inform Huawei's inference infrastructure strategy.
- Proposed model routing and tiered KV-cache storage as research directions ahead of their industry mainstreaming; helped pivot the team's research focus toward agentic AI serving.
- Initiated and designed an agentic AI hackathon on openEuler OS in partnership with the University of Edinburgh, authoring all challenge problems and attracting 100+ student participants.

PROJECT EXPERIENCE

Personalized Dialogue Generation with Implicit Persona Modeling

NLP, In-Context Learning, Dialogue Systems

Mar 2024 – Sep 2024

UCL MSc Dissertation

- Engineered an in-context learning pipeline that inferred user personas from dialogue history alone rather than from predefined profiles; outperformed most existing implicit- and explicit-persona systems on fluency and diversity across GPT-3.5/4o and six open LLMs.
- Proposed Average Drift Score, a turn-based metric quantifying topic drift; showed that drift tendency is consistent within speakers and contagious across conversation partners.

Scientific Data Compression with Variational Autoencoder

PyTorch, VAE, Entropy Coding

Oct 2022 – Jun 2023

UoL BEng Dissertation

- Adapted a scale-hyperprior variational autoencoder for lossy compression of 3D seismic waveform data, achieving lower-bitrate operating points than the ZFP baseline at matched distortion.

EDUCATION

University College London

MSc Integrated Machine Learning Systems – *Distinction*

London, UK

Sep 2023 – Sep 2024

University of Liverpool

BEng Computer Science & Electronic Engineering – *First Class Honours*

Liverpool, UK

Sep 2020 – Jul 2023

SKILLS

ML & AI: LLM Inference Optimization (batching, prefix caching, KV-cache management), Agentic AI Serving, PyTorch
Systems & Infrastructure: vLLM, vLLM-Ascend, CrewAI, OpenTelemetry, Prometheus, Docker, Kubernetes
Hardware: Ascend NPU/CANN, NVIDIA GPU, Cerebras wafer-scale, memory and interconnects (HBM, HCCS, CPO)
Programming: Python, C++, Java, SQL, Linux/Unix Shell, Git
Languages: Chinese (Native), English (Fluent), Japanese (Conversational)
Interests: Hardware architecture deep dives, AGI and post-AGI futures, equity and options trading